Asfand Yar | GenAI Backend Engineer

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AI-First Engineer with hands-on experience building intelligent applications, backend services, and scalable APIs using Python and FastAPI. Skilled in Generative AI, developing RAG pipelines, working with open-source LLMs, and applying prompt engineering to improve application performance. Experienced in building Agentic AI solutions, integrating AI capabilities into backend workflows, and containerizing applications using Docker. Focused on designing reliable, efficient, and production-ready systems that connect generative AI with practical business applications.

Experience

Emumba — **Onsite - Islamabad**
GenAI Engineer Intern — *Jan 2026 – Mar 2026*
- Designed scalable backend systems using Python and FastAPI, integrating Generative AI capabilities and prompt engineering to optimize LLM outputs for enterprise applications.
- Engineered Retrieval-Augmented Generation (RAG) pipelines and agentic workflows to improve contextual responses, knowledge retrieval, and AI-driven application functionality.
- Collaborated with cross-functional teams to integrate AI agents, APIs, and backend services, ensuring efficient, low-latency data processing and seamless deployment of AI-powered features.

Developers Hub Corporation — **Remote**
AI Intern — *Mar 2025 – May 2025*
- Engineered Python scripts to extract, clean, and format raw data, creating high-quality datasets for testing internal machine learning models.
- Conducted foundational prompt engineering and benchmarked various Open-source language models to document performance accuracy.
- Built internal UI dashboards and automated scripts to visualize AI outputs and test API endpoints, streamlining day-to-day development tasks.

Technical Skills

- AI & Machine Learning: Generative AI, Large Language Models (LLM), RAG Pipelines, Prompt Engineering, AI Agents, AI Automation
- Frameworks & Tools: FastAPI, LangChain, OpenAI API, n8n, Pipecat, Cartesia, Github, Git
- Languages & Development: Python, SQL, Node.js, REST APIs, HTML5, CSS3, React.js, Tailwind CSS
- Data & Infrastructure: Redis, Docker, Containerization, Virtual Environments, Vector Databases

Key Projects

Real-Time Audio & RAG
AI-Powered Voice Bot
- Implemented a real-time conversational voice agent using a local LLM, Pipecat, and Cartesia for natural text-to-speech interactions.
- Engineered a RAG pipeline with a vector database to enable the voice bot to answer complex, context-specific queries accurately.
- Managed the project in a virtual environment, containerized the application using Docker, and successfully published the image to Docker Hub for public access.

FYP Project
Intervexa - AI Web-Based Mock Interview System
- Developed an AI-driven interview platform that dynamically tailors technical questions based on the candidate's specific job role and skill set.
- Integrated voice recognition to analyze verbal responses and facial expression capture to monitor candidate confidence during the interview.
- Provided a comprehensive feedback dashboard to help users prepare for real-world interviews effectively.

National AI Hackathon
Decision Debater AI
- Built an AI-Powered debate platform using LangChain and Retrieval-Augmented Generation (RAG) to provide evidence-based arguments.
- Integrated vector search with LLM to retrieve relevant information and generate context-aware responses from multiple knowledge sources.
- Created backend APIs using FastAPI to support real-time debate generation and seamless integration with frontend applications.

Education

BUITEMS — **Quetta**
Bachelor of Science in Computer Science, GPA: 3.83/4.00 — *Oct 2022 – Dec 2026 (Expected)*

Tameer-e-Nau Public College — **Quetta**
Intermediate in Pre-Engineering, Grade: A+ — *Mar 2019 – Apr 2021*

Certifications & Workshops

- ACT AI National Training Programme (2026)
- Microsoft Learn Student Ambassador (2024)
- Generative AI Application Developer Certificate
- ChatGPT for Everyone (Learn Prompting)
- Google AI Essentials (2024)
- National AI Hackathon (atomcamp)
- Anthropic Claude Code in Action
- WEB3 Masterclass (Algorand)

Leadership & Achievements

BUITEMS Entrepreneurial Society — **Quetta**
Executive Lead — *Aug 2024 – Present*

Mobile Phone Repairing & Maintenance — **EME Center, Quetta Cantt**
Gold Medalist — *2020*

Languages

English: Advanced | Urdu: Professional | Punjabi: Fluent | Pashto: Intermediate